



Creo Assessment

Computer Aided Engineering (MHH113285) - PTC Creo Major Assessment

(Bilal Shabbir - S2324519)

Dr Muditha Kulatunga

Dr Andrew Cowell

Mr David Semple

Dr Sheng Cheng

This piece of coursework is my own original work and has not been submitted elsewhere in fulfilment of the requirement of this or any other award

Table Of Contents

Introduction	<i>page 3</i>
Background Research	<i>page 4</i>
Initial Ideation	<i>page 5</i>
Part Design in Creo	<i>page 6</i>
Working Drawing	<i>page 9</i>
Mould Design and Analysis	<i>page 10</i>
CNC Machining of Mould Half	<i>page 18</i>
Injection Moulding	<i>page 23</i>
3-Axis CNC	<i>page 24</i>
Conclusion	<i>page 25</i>
Appendix	<i>page 26</i>

Introduction

This report documents the computer-aided engineering work used in the design and manufacture of a moulded plastic housing component for a domestic hair dryer using PTC Creo. The aim is to develop a robust 3D part model suitable for high-volume injection moulding, generate an appropriate mould design, and create a machining strategy for one half of the mould using 3-axis CNC techniques.

The work begins with the modelling of half of a hair dryer body as assigned, focusing on capturing the key geometric features while ensuring that design for manufacture principles, such as draft angles, wall thickness, and the provision of a suitable parting plane, are respected. The model is then used to generate a two-part injection mould, including the parting surface and runner system, and to perform mould analysis in Creo to assess fill behaviour and overall mouldability of the part. Subsequently, one half of the mould is prepared for manufacture using Creo's machining module, where toolpaths, machining parameters, and NC codes are generated and reviewed.

The report is structured as follows. First, the part design steps and associated reflections are presented. This is followed by mould design and mould analysis, with commentary on the mouldability of the part and any required design modifications. The subsequent section describes the machining strategy for one half of the mould, including sample NC code. Finally, a concise overview of general injection moulding and 3-axis CNC machining processes is provided, supported by appropriate references, before concluding remarks.

Background Research

This section provides an overview of the injection moulding and CNC machining processes to place the hair dryer and mould design work in an appropriate manufacturing context. By researching established industrial practice, the report ensures that the subsequent CAD modelling, mould design, and machining methods are informed and use the best-practice guidelines.

In order to design a product that is successful in its function and can be mass-manufactured, a specification had to be created. This specification must be based on research from various sources.

Manufacturing Specification

Crescent industries <appendix a> Crescent Industries, 2023

Crescent Industries published a journal containing best practices for designing injection-moulded products. These include:

1. Part Geometry - The design must include a flat plane for the parting surface of the model.
2. Wall Thickness - having too thick or varying wall thicknesses can cause air traps, warping, and incomplete filling.
3. Draft Angles - The designed part must be able to be ejected from the mould easily without sticking, scraping, or damaging the surface

4. Initial Ideation

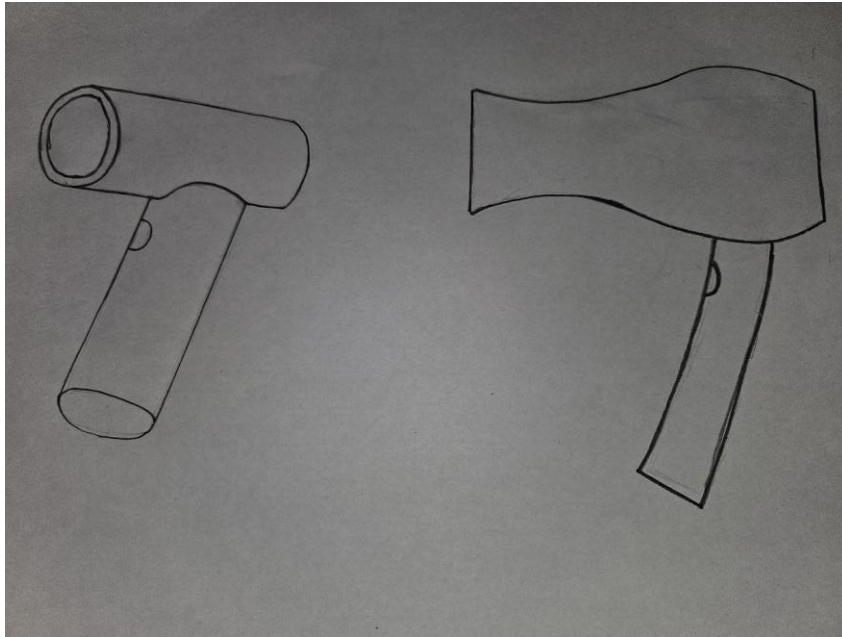


Figure 1.1 Two early design options for the hair dryer housing. Option 1 uses simple cylindrical geometry, while Option 2 incorporates a choke shape and ergonomic handle, providing higher airflow and improved usability.

Two Designs were created: option 1 (isometric), shown on the left, and option 2 (orthographic elevation), shown on the right.

Breakdown

Option 1 is a modernist take and would be simple to model and manufacture due to its shape being two cylinders; however, with no choke bottle shape, there is a low amount of maximum airflow in comparison. Option two accelerates the air into a high-pressure zone before exiting through a larger shaped nozzle.

Furthermore, the handle on the second option was more ergonomic and would feel better to use regularly.

In this situation, option 2 was chosen to be taken forward to the modelling phase due to the higher air flow and more ergonomic design.

Part Design in Creo

This section describes the creation of the CAD model of a hair dryer housing using Creo. The component forms the outer shell of a hair dryer and is designed to enclose the internal electrical and mechanical components while providing a comfortable grip for the user. The model was developed with the intention of being suitable for injection Moulding, so attention was given to overall form, wall thickness, draft angles, and the provision of a clean parting plane. The key external features include the main body and the handle.

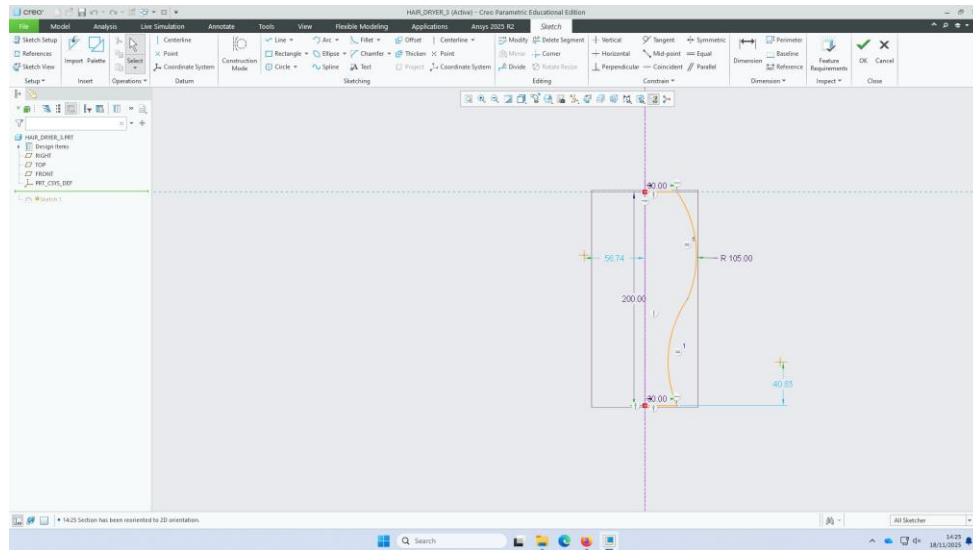


Figure 2.1 The first profile used to generate the top body of the hair dryer. This defines the main external curve prior to revolving.

Shown above is the shape that was revolved to achieve the top body part. A solid revolve was used, as for a shell to be used, the full part was created.

After revolving, a sweep was used to create the curved handle section, and a cutout was created for the switch to be placed in. This was done by offsetting a plane to the desired height and drawing a sketch on it, as shown in the figures below. The sketch was then extrude-cut in both directions to create the opening for the switch.

Furthermore, the edges were filleted, and a pattern was added to the back side to allow airflow through the dryer.

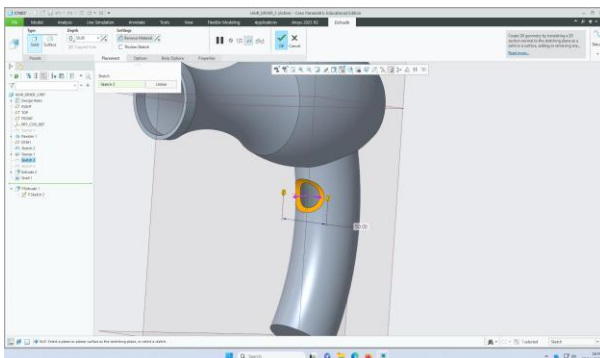


Figure 2.2 Extruded cut added to form the switch opening on the handle area, created using an offset plane.

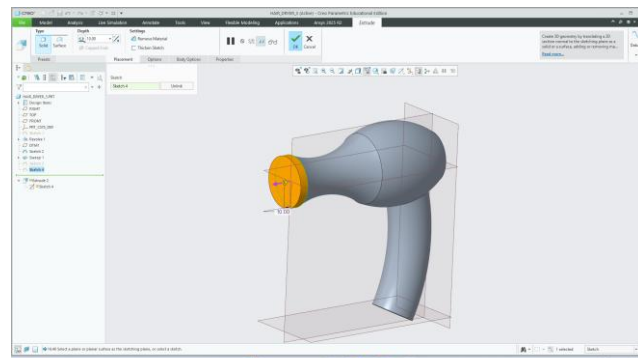


Figure 2.3 Extrusion defining the nozzle geometry, shaping the air outlet of the hair dryer

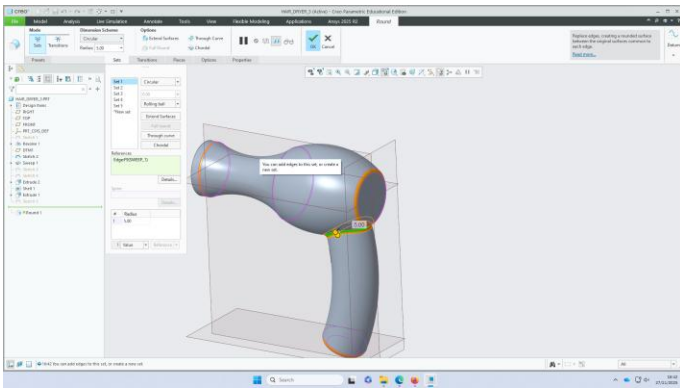


Figure 2.4 Application of fillets to smooth sharp transitions, improving both ergonomics and mouldability.

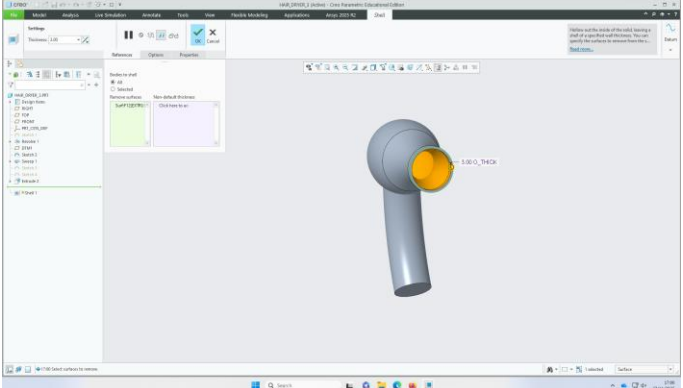


Figure 2.5 The shell feature used to hollow the part with a uniform wall thickness suitable for injection moulding.

An extrude cut was used to have the model after a shell of wall thickness 3mm was done. The sweep was used with an offset plane outside of the body and then extruded up.

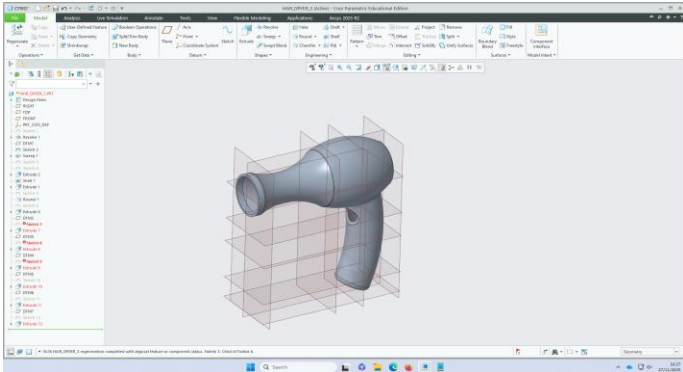


Figure 2.6 Completed solid revolve that forms the main hair dryer body prior to internal detail.

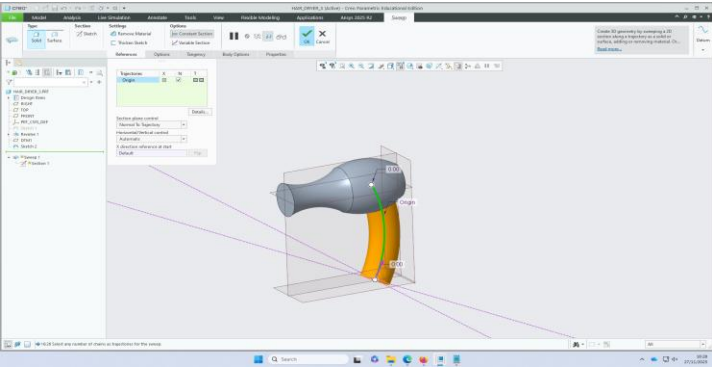


Figure 2.7 Sweep feature used to generate the curved handle.

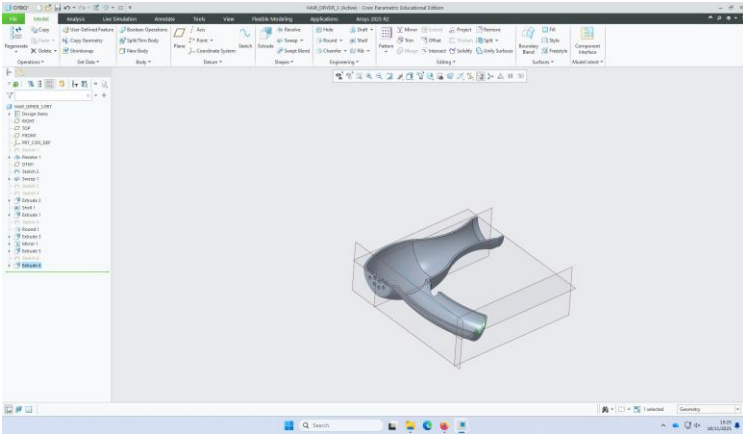


Figure 2.8 The body after being halved to allow access for internal webbing.

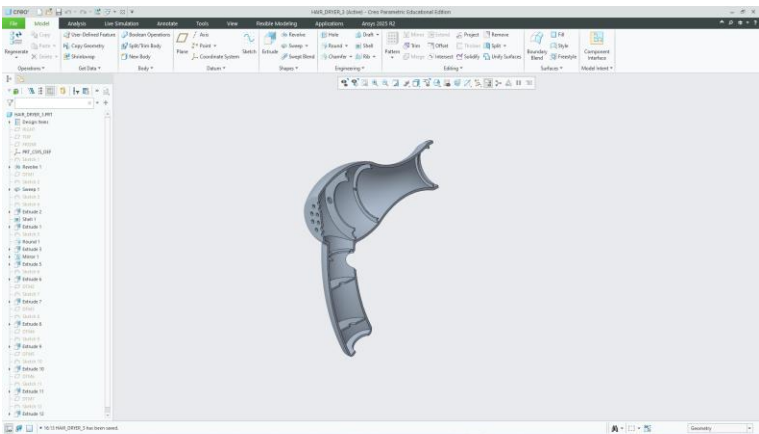


Figure 2.9 The final internal pattern and web structure supporting the motor, heater, and switch

After the webs and mounting hardware were added to the internal structure, the final model was finished and ready for the mould to be created.

This was done using the web tool, which added a repeatable structure in the hair dryer.

Two webs with mounting holes were added to hold the motor in place, and the fan would be held in by the motor. Furthermore, there were also two webs for the heating component. A mounting point in the bottom was added to support the power cable, and two for the electronic switch in the handle were also used.

This involved creating many offset planes and using the web tool on each offset plane, then setting a thickness of 5mm.

A 2D Working Drawing was then created with the final design, with some minor tweaks to make the design easily mass manufacturable.

The vent holes in the back were removed as they would allow for the part to be ejected, and the mounting holes for the electrical components were also removed; these could be added later through an additional manufacturing process.

Problems encountered

Initially, a surface revolve was used, then a thickening; however, this would result in an uneven wall thickness when fillets are added in. Therefore, a solid revolve should have been used; this proved to be difficult as Creo wouldn't recognise the centerline as an axis for a solid revolve. The shape was remade as a connected shape and then revolved around its centre. The handle was created using a sweep; however, the tool wouldn't work on the offset plane, which had been created below the upper body. A solution used was to offset the plane inside the top body slice the graphics and extrude from there downwards.

Working Drawing

As you can see below, the working drawing has all major views, each with the major dimensions necessary for manufacturing. Furthermore, the end vent holes were removed as these would be punched out after the injection moulding process.

A detailed view of the major web holding the fan in place was also created, along with an isometric view in the top right.

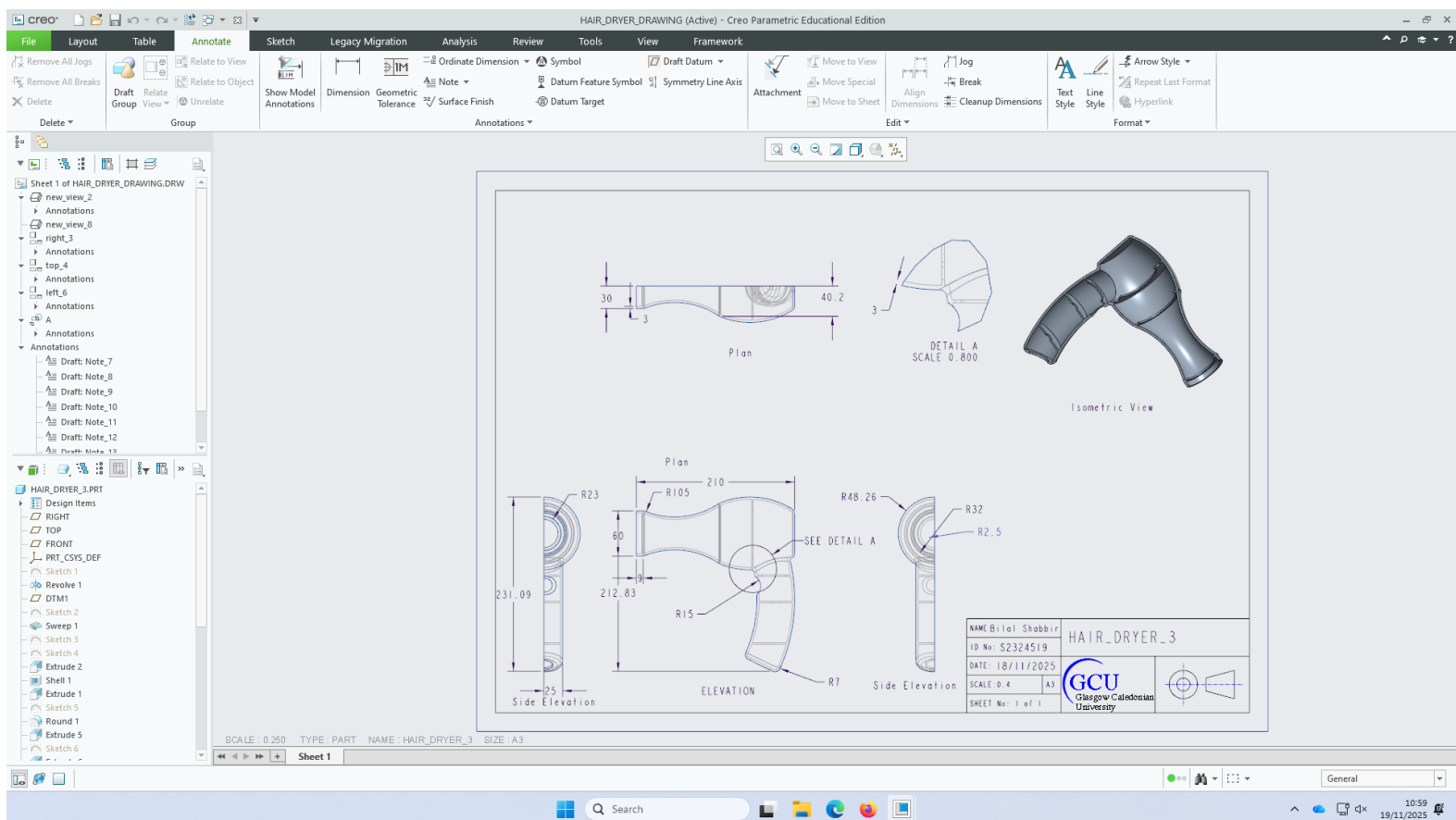


Figure 2.9.1 Orthographic views, dimensions, and an isometric view of the final part prepared for manufacture

Mould Design and Analysis

This section describes the development and evaluation of the injection mould required to manufacture the hair dryer housing component. Using the completed part model as input, a suitable parting direction and parting surface are established, followed by the design of the mould cavity, core, and runner system within Creo. A mould filling analysis is then performed to assess the flow behaviour of the molten polymer, identify potential issues such as air traps, weld lines, and unbalanced filling, and evaluate overall mouldability.

When designing components like the hair dryer body, it is important to consider that all plastics shrink slightly as they cool in the mold.

ABS has a shrink rate of around 0.5-0.7% <Appendix F>, which is relatively low compared to other plastics, meaning it has an even thickness. Giving consistent wall thickness helps to make sure that the part shrinks evenly, reducing the risk of warping or internal stress. The mold cavity must be slightly bigger to compensate for the shrinking so that the final part meets the correct dimensions. Smooth transitions between curves and not having any sudden changes in thickness helps to have tight tolerances and improves the finish.

First, the design must be prepared for injection moulding, i.e., the mounting and vent holes must be removed as they would stop the ejection of the part from the mould. The now prepared part was placed in a new file for the mould, and a rectangular shape was extruded around the part. This would become the mould.

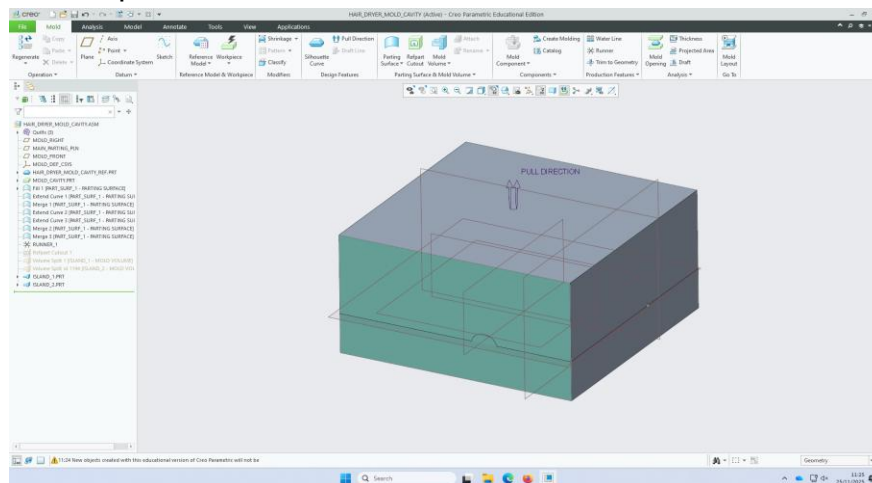


Figure 3.1 Initial mould block created to represent the cavity and core surrounding the hair dryer housing.

After this, a parting surface had to be created, and this proved to be a big challenge as the tool to create an extended curve would not allow for a full parting surface around the curves in the handle.

To solve this, the fill tool was used, and references of the full corners of the mould limits and the part after this, the extend curve tool was used on just the curves, and the limits were set at the end of the mould. After this, all the surfaces were merged into one, allowing for a full parting surface.

The parting plane was positioned along the mid-plane of the hair dryer as this provided the most practical location for splitting the mould. Placing the parting line here ensures that most of the external faces draft naturally in the opening direction, which helps avoid undercuts and reduces the risk of the part sticking during ejection. The mid-plane also follows the external symmetry of the component, meaning halves remain relatively simple to machine using standard 3-axis CNC operations. By avoiding complex changes in curvature or recessed areas, this parting plane helps maintain a clean and continuous split line around the entire component. Overall, choosing this location improves mouldability, simplifies machining, and results in a more reliable mould design.

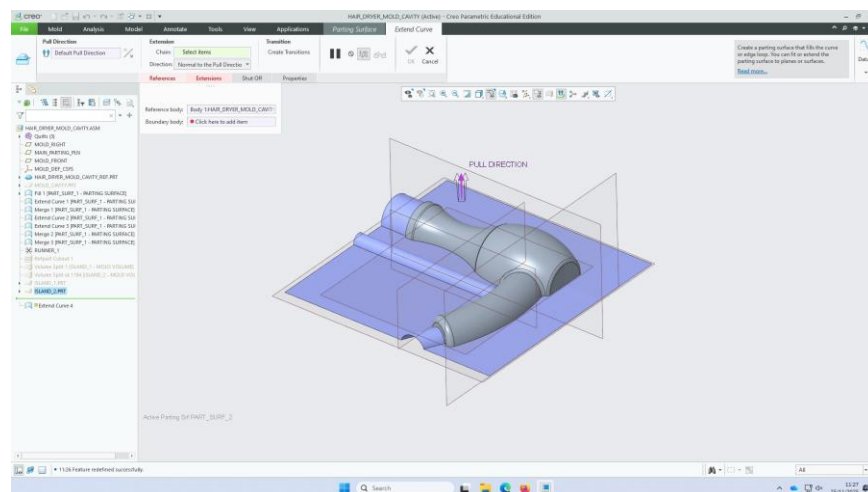


Figure 3.2 Generated parting surface ensuring clean mould separation.

After this, the runners were added, and the mould was split open. The runner's location is important, as this can impact on the air traps and the quality of the part.

The runner diameter was 5mm as it provided a consistent flow of material to each gate while stopping the plastic from cooling too quickly as it travels through the mould. A small runner can slow the flow of plastic, whereas an oversized runner wastes plastic and increases the time it takes to mould. 5mm is a compromise by giving a steady flow rate, lowering pressure loss along the runner, and helping to make sure that each gate gets an even supply of molten plastic. This improves the overall mouldability.

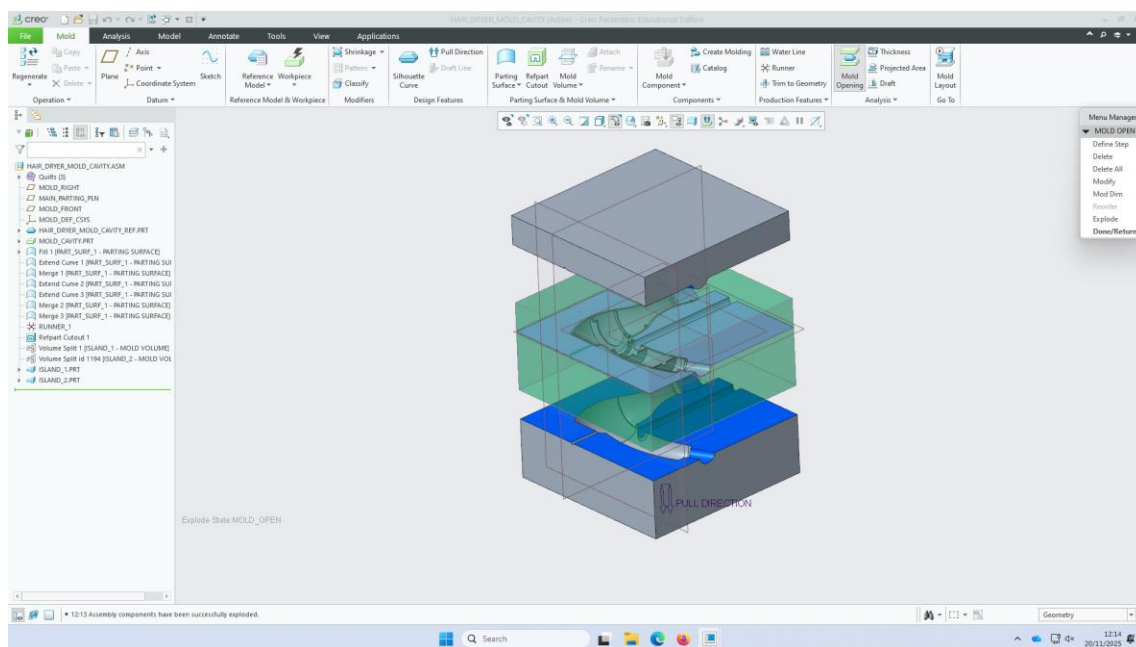


Figure 3.3 Section showing how the mould splits with the runner system.

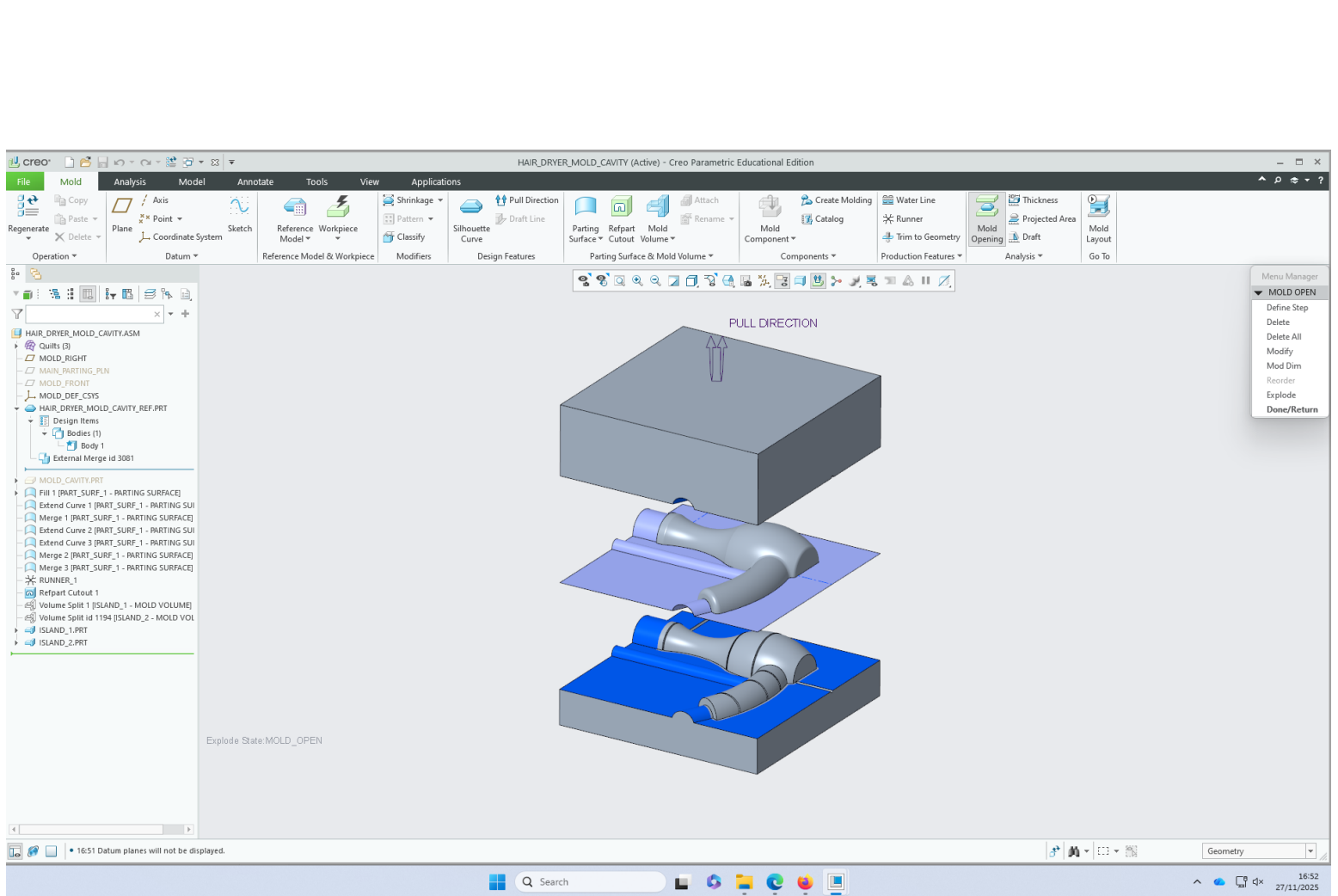


Figure 3.4 Final separated blocks after the runner system and parting surfaces have been applied.

Once the runner position had been tweaked and adjusted, the part was moved onto mould analysis, where the flow of the plastic would be analysed. The first part of this process was to choose a plastic type. The two primary options were:

- A PC-ABS Blend - Due to its high flow rate and low cooling time
- A DuPont Zytel - Due to its high strength and it absorbs moisture

Property / Consideration	PC-ABS	Zytel (Nylon 66)
Stiffness & Strength	Good strength more rigid than ABS	Very high strength and stiffness higher fatigue resistance
Surface Finish	Smooth surface finish	More difficult to achieve smooth finish often textured
Moulding Behaviour	Easy to mould and forgiving, suitable for complex housings	Requires higher melt temperature can be harder to fill thin sections
Suitability for This Part	Well suited for the hair dryer housing due to good stability and finish	Less suitable because of shrinkage and moisture absorption which may distort thin walls

After a short comparison, a PC - ABS blend, specifically BAYBLEND FR 3010, has a high heat resistance, which is important as the hair dryer has a heating element, and good flow during the injection moulding process, which would help make sure the part is created correctly.

After this, the gates were added to the model. At first, only two gates were added, each at a diameter of 5mm, as shown.

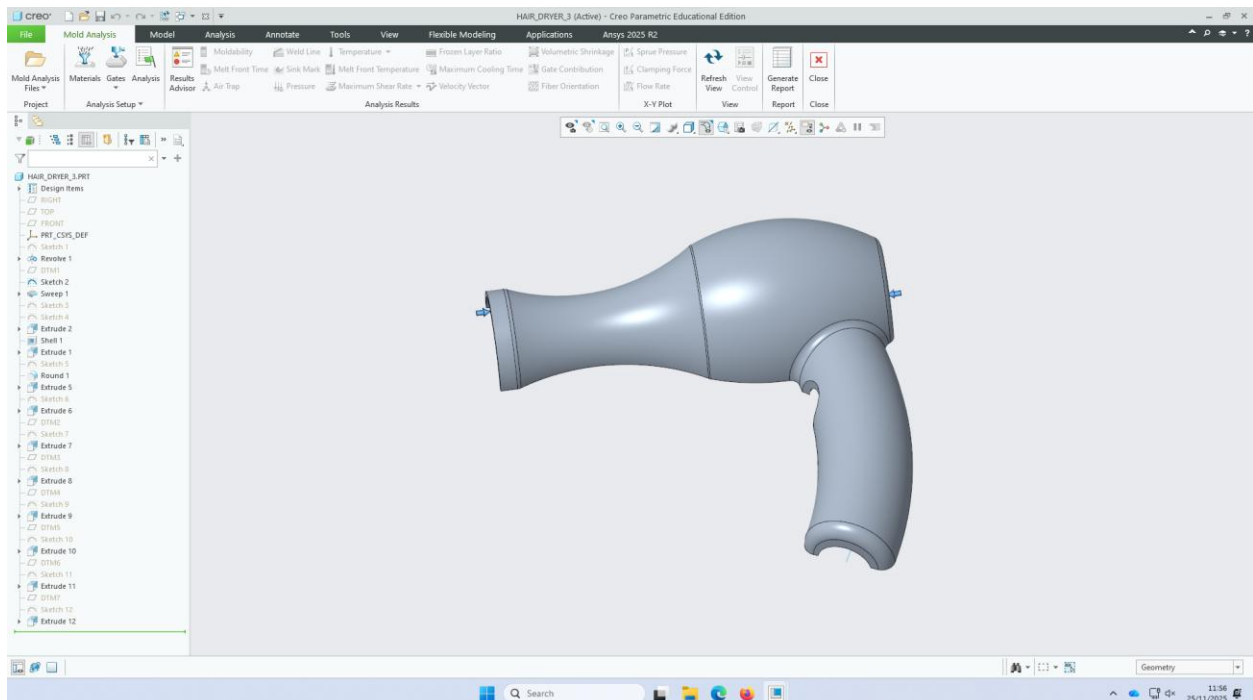


Figure 3.5 The first gate configuration using two gates at 5 mm diameter

The Model was analysed, and the results are shown below.

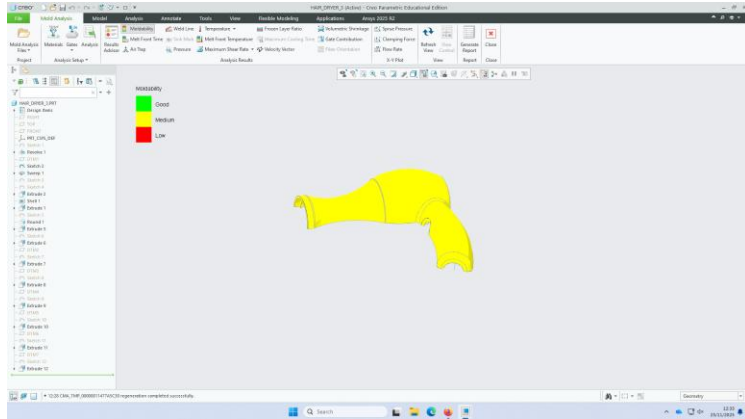


Figure 3.6 Early mould analysis showing areas of poor mouldability.

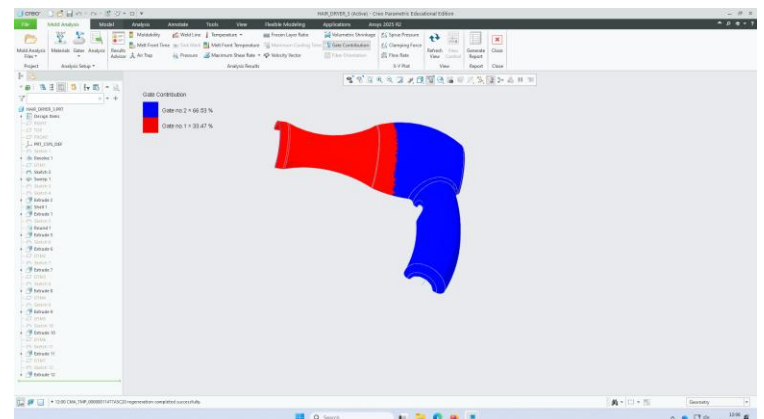


Figure 3.7 Flow imbalance showing high flow from a single gate.

As you can see, the mouldability is not good, the gate contribution is also heavily skewed to fix this, and the large differences in temperature in the moulding process between different parts of the model, it was concluded that a third gate should be added.

A third gate was added, as shown below which reduced the air traps, helped the mouldability, and evened out the gate contribution.



Figure 3.8 Revised mould analysis showing much better mouldability.

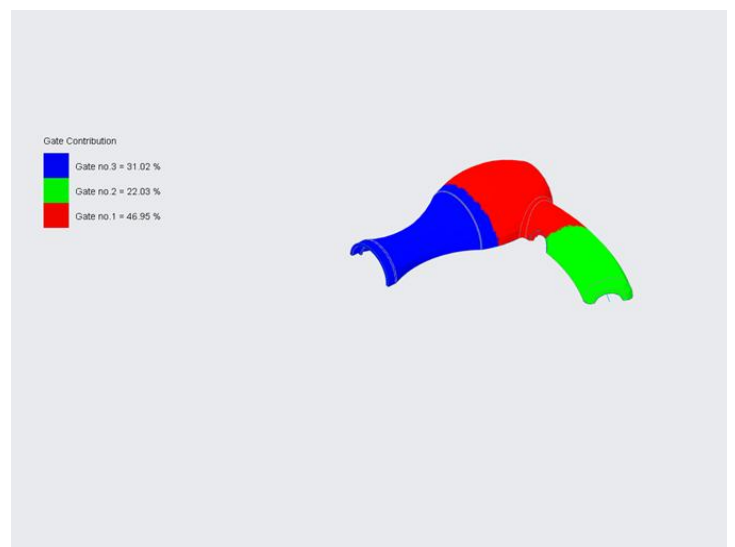


Figure 3.9 Balanced gating layout, with each gate contributing more evenly to the filling.

As you can see, the mouldability is good all over, which is due to the addition of a third gate and an increase in the melt temperature.

Additionally shown below are images of the air traps and the flow rate graph.

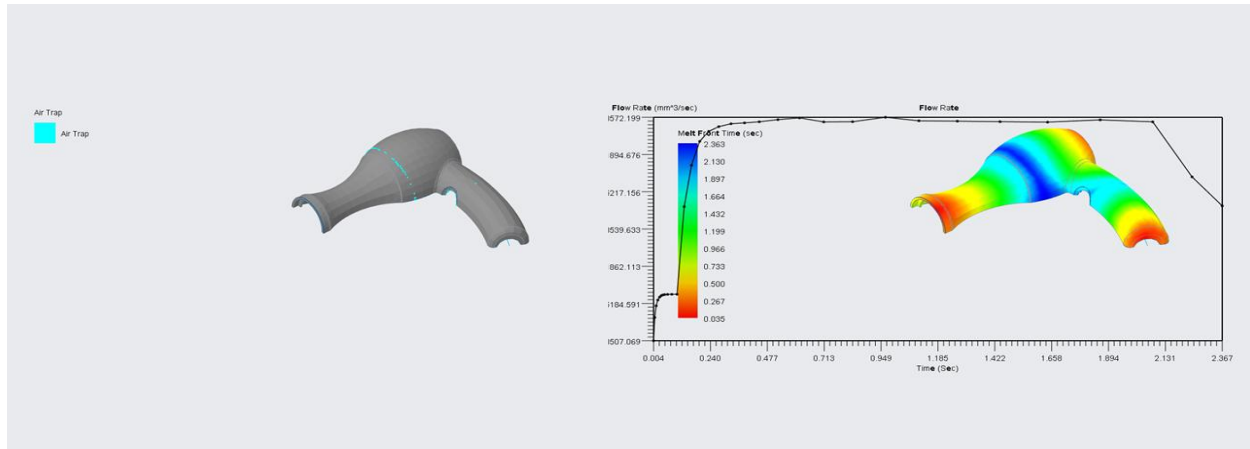


Figure 3.9.1 Simulation output highlighting likely air traps

Figure 3.9.2 Flow rate plot showing improved flow.

Also shown below are the maximum shear stress results and weld lines.

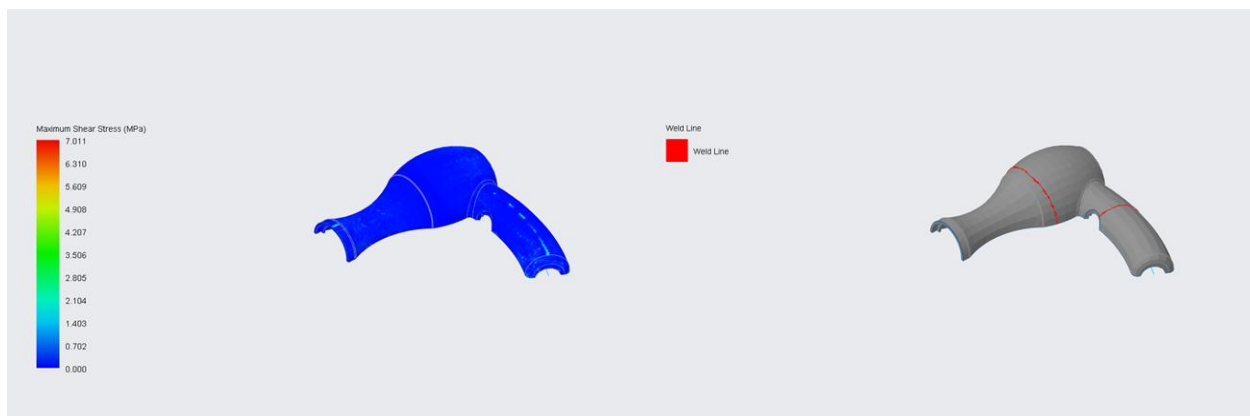


Figure 3.9.3 Shear stress distribution showing that peak stresses are low enough for manufacture.

Figure 3.9.4 Weld indicating surface regions where fronts meet.

As shown, the shear stress and flow rate were all as expected and sufficient for the part to be taken forward and moulded.

However, the air traps and weld lines are not; a solution must be designed.

This is because having the weld lines can cause surface defects, and they could be visible.

Air traps also cause surface defects and can cause voids or serious defects.

Solutions to make the weld lines less visible could be to:

- Increase the mold temperature or melt temperature.
- Decrease the filling time.
- Reduce the number of gates.
- Use fiber-reinforced or low viscosity material.
- Increase injection speed

Solutions to reduce the air traps

- Increase the filling time.
- Change the gate location.
- Reduce the injection speed.
- Modify the size or location of the vents.

A final manufacturability check showed that all important surfaces had at least 1° of draft, the wall thickness stayed consistent at 3 mm, and there were no undercuts that would require side actions. This helps make the mould easier to machine and ensures the part can be ejected smoothly without sticking.

In Summary, a combined solution would be to reduce the injection speed and increase the melt temperature. This would still result in having some weld lines; however, they were within tolerance for manufacture, and the same can be said for the air traps.

CNC Creo Machining of Mould Half

This section outlines the process of generating the CNC machining strategy for one half of the injection mould using the manufacturing tools within Creo. The aim is to prepare a machinable mould block by defining the workpiece, establishing an appropriate coordinate system, and developing a sequence of toolpaths suitable for 3-axis CNC production. This starts with setting up the manufacturing model, where the stock material, orientation and datum references are specified. This initial setup provides the foundation for creating subsequent operations such as facing, roughing and finishing, and ensures that all toolpaths are aligned correctly with the chosen machining direction. The remainder of the section then presents the machining operations, cutting parameters, toolpath simulations, and sample NC code, followed by a brief reflection on any issues encountered during the process.

This process was started by creating a new type of file and placing the negative half of the mould in it. After this, a sketch was created on the bottom plane, and a rectangle was extruded up, extending 10mm beyond the top of the part to allow for the mill to drill properly.

A coordinate system was then added so the CNC machine knew the orientation of the X, Y and Z axes. The reference was set at the corner of the mould half, with the X and Y directions flipped as shown

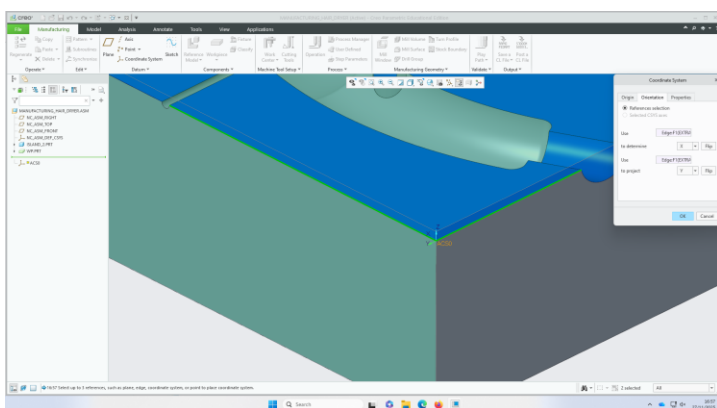


Figure 4.1 Detailed view of the coordinate system defining the origin and tool orientation.

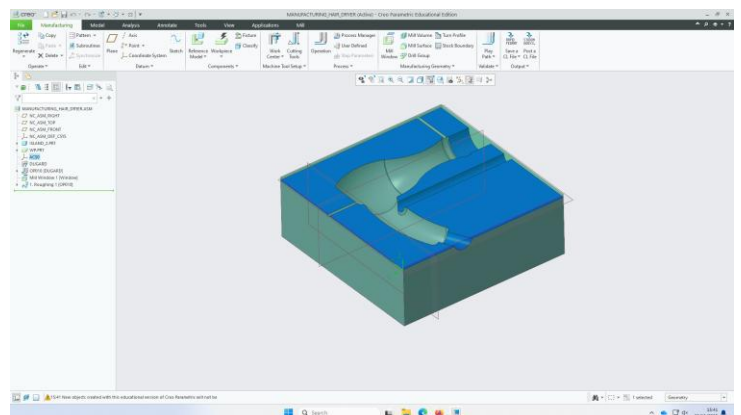


Figure 4.2 Full mould block showing the coordinate system used for CNC programming in Creo.

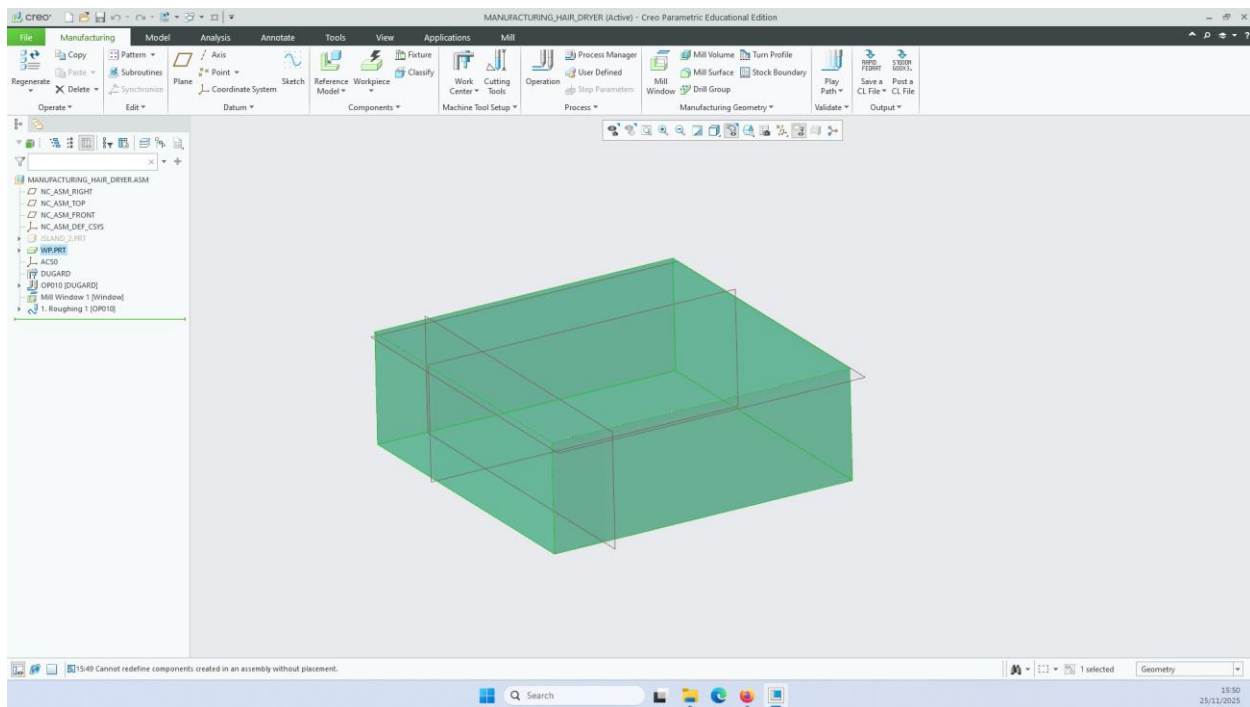


Figure 4.3 Stock material extended 10 mm above the mould to allow safe initial drilling and roughing.

After this the milling tool centre was set up with the number of axis set to 3 and the name to Duguard, CNC control to Siemens then the parameters were set:

- RPM = 2000
- HP = 5

A spindle speed of 2000 rpm was selected as it provides a suitable balance between material removal rate, surface finish, and tool life when machining aluminium mould tooling on a 3-axis CNC machine.

A step-over of 2 mm was used for finishing to achieve a smooth surface

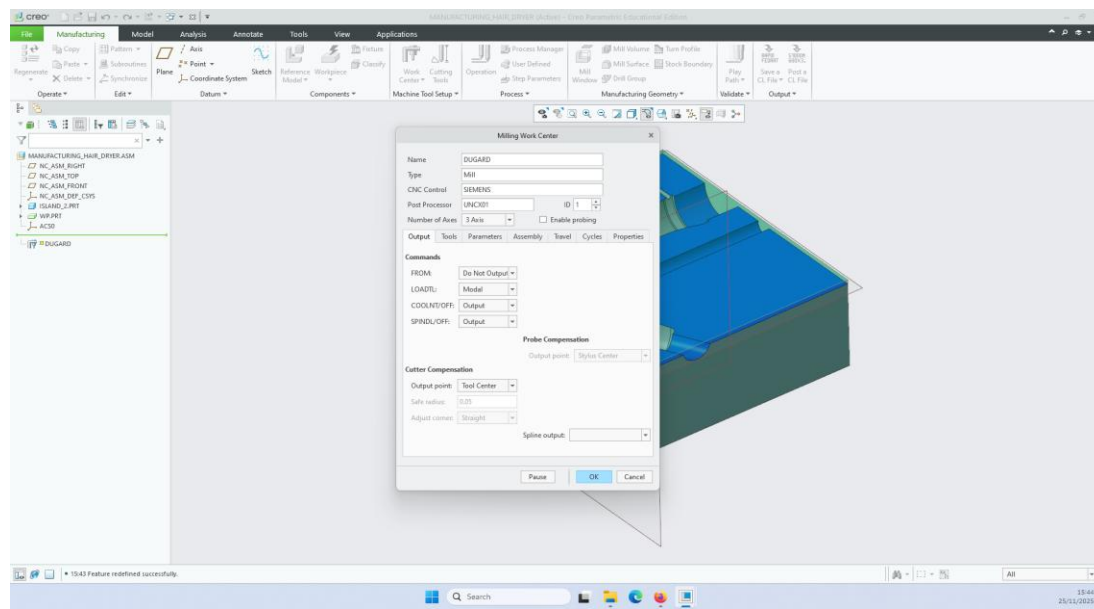


Figure 4.4 Tool configuration including cutter diameter, spindle speed, and operational parameters.

After this, the tool was set up with the diameter and length of the roughing bit. This is vital to achieve a balance between precision and machining time. Additionally, a finishing tool was added over the more difficult components, and this was used a smaller bit to be more precise.

Then a milling window was added, and the milling process was also added. Then the analysis was run and compared with a different colour representing the stock and the part.

The Milling window was sketched on the top surface of the extrusion as the mill should start there to avoid damaging the part instead it should start on the sacrificial material. Furthermore, the tool was made in the tool manager where the parameters and diameter of the roughing tool were made.

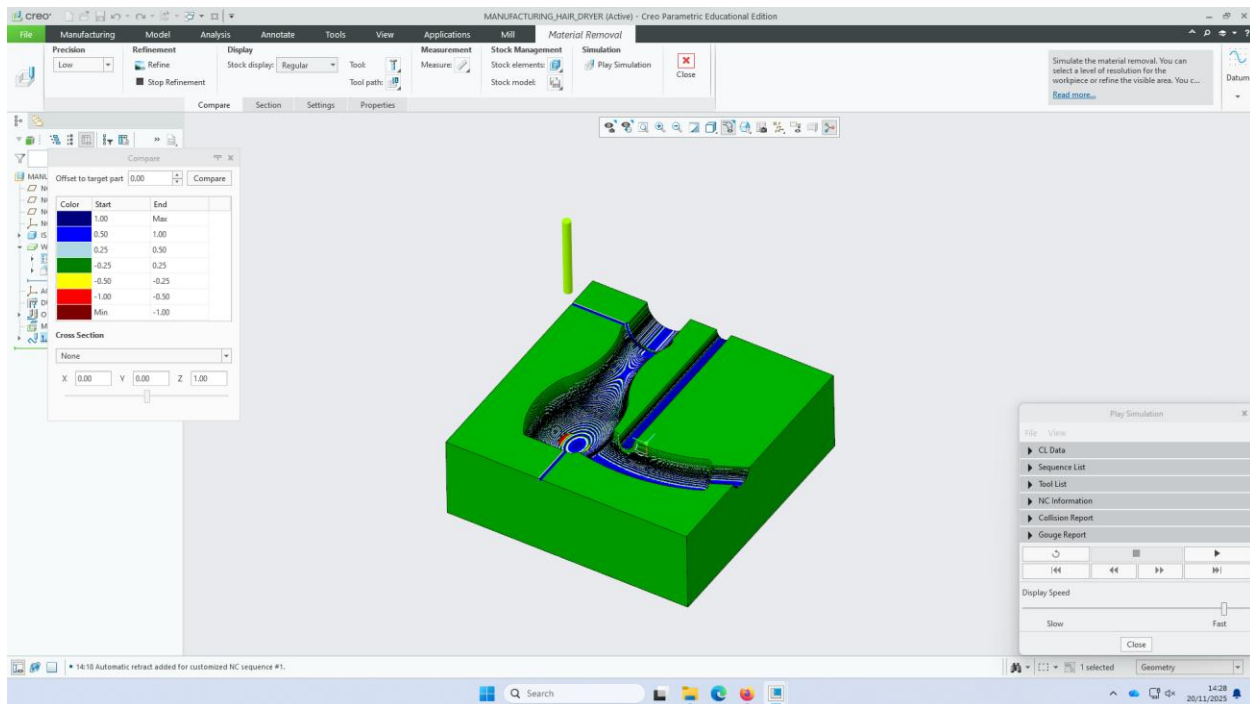


Figure 4.5 Initial roughing pass leaving significant uncut material due to large tool diameter.

As you can see, the drill left a large amount of material on the mould half and drilled too much in the centre of the hair dryer. This was done with an initial bit diameter of 10mm. The machining time was low, but it must be made smaller to make the mould viable. The analysis was run again with a smaller bit size and a finishing tool.

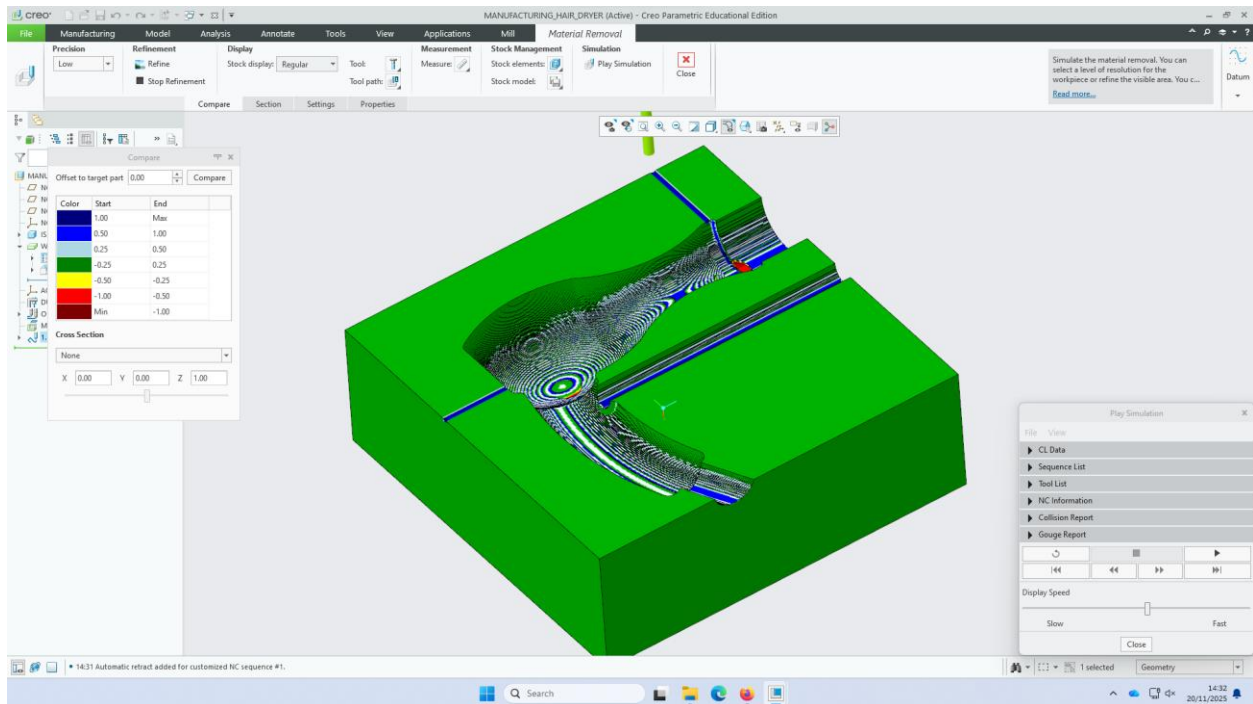


Figure 4.6 Refined roughing pass providing better accuracy and reducing leftover stock.

This attempt was much better; however, it was still not precise enough. The roughing was small enough; however, the finishing would need to be made smaller, close to 4 mm. The last refinement uses a roughing bit size of 4mm and a finishing size of 2mm.

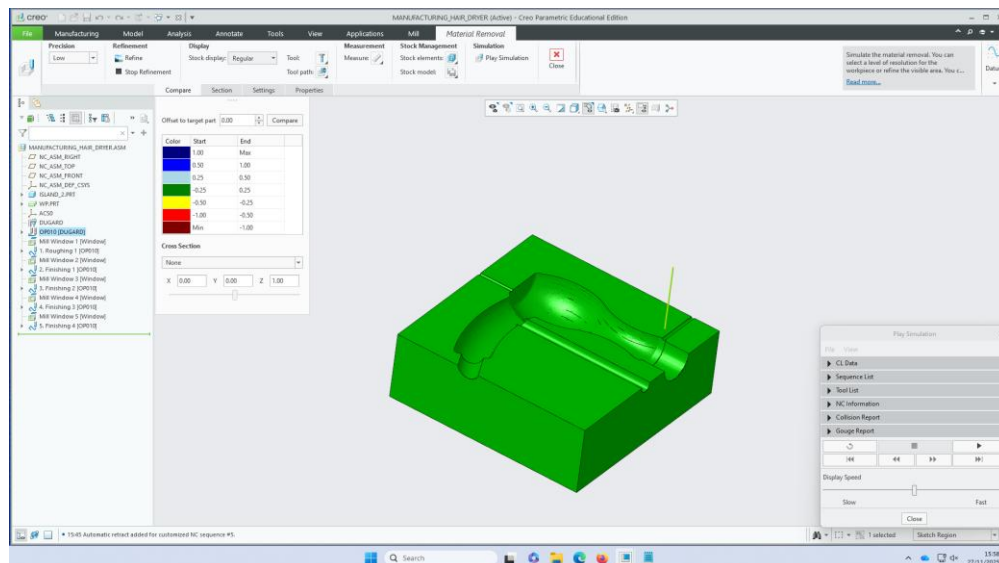


Figure 4.7 Optimised toolpath set achieving accurate geometry and smooth surfaces.

The Finishing required new milling windows to be added this was done by sketching with reference to the edges of the part.

This was deemed to be sufficiently refined as no excess material was left, and no deformities were left on the part.

The first 40 lines of the G-code will be placed in the appendix <appendix b>

Operation	Tool Diameter	RPM	Step-over	Notes
Roughing (Attempt 1)	10 mm	2000 rpm	—	Too large; left excess stock
Roughing (Final)	4 mm	2000 rpm	—	Good removal rate
Finishing	2 mm	2000 rpm	2 mm	Achieved smooth cavity finish

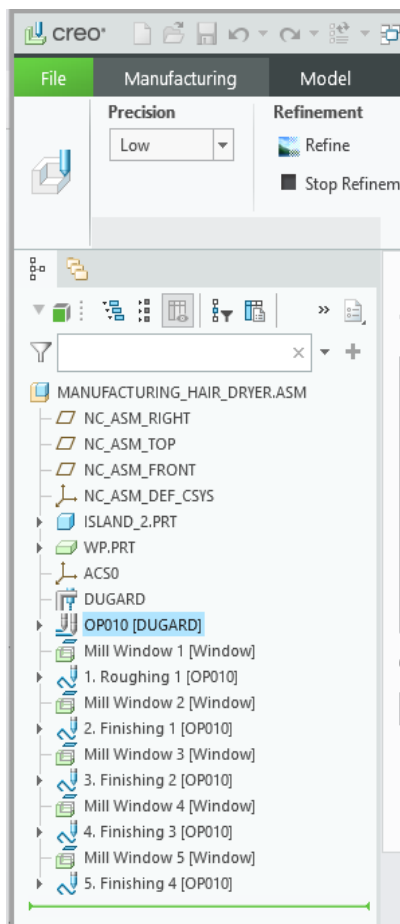


Figure 4.8 Process tree showing the full sequence of operations performed in Creo.

Manufacturing Processes

This section gives a explanation of the injection moulding process and the 3-axis CNC machining process, both are related to how the hair dryer would be moulded and how the mould would be created. Understanding how these processes work is important, as it helps to make sure that the part and mould are designed in a way that is realistic, can be manufactured easily, and follows the standard industry practices.

Injection Moulding

Injection moulding is a common manufacturing method used to produce plastic components in large quantities. The process starts with plastic granules being poured into a hopper which is heated, after this they are melted and mixed by a corkscrew. Once the correct melt temperature and pressure are reached, the screw pushes the molten plastic into a closed mould at a high speed and high pressure.

After the mould is filled, the plastic is held at a high pressure while it cools. Cooling vents built into the mould help dissipate heat so that the part cools evenly. Once cool, the mould opens, and the ejector pins push the part out, and then the mould is ready for the next cycle. This process repeats quickly, which is why injection moulding is most often used for mass production.

For a part to be moulded successfully, it must include features such as uniform wall thickness, draft angles to allow ejection, and a parting surface that lets the mould split cleanly. If these are not considered, issues such as warping, air traps, weld lines, or incomplete filling can occur. <Appendix C> <Appendix D>

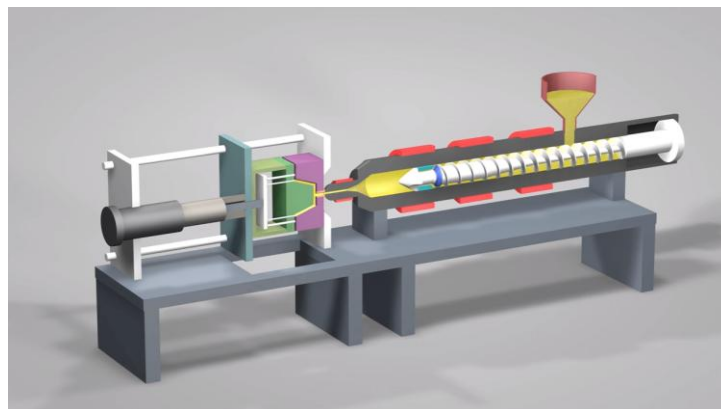


Figure 5.1 Diagram showing hopper, screw, heater, and injection into a closed mould.

3-Axis CNC Machining

3-axis CNC machining is a subtractive process used to remove material from a solid metal block to create parts, such as the cavity of a mould. A CNC milling machine operates along X, Y, and Z axes, allowing the tool to cut, and drill the workpiece.

The setup begins by fixing the stock material to the machine table and defining a coordinate system, this is done in Creo, so the machine knows where the origin is. Toolpaths are then created to guide the cutting tools through roughing, and finishing passes. Roughing removes the majority of the material, while finishing produces the final surfaces and fine details. This is done as the roughing tool is wide and meant to be used to get rid of material quickly whereas the finishing tool will be smaller and therefore slower.

Tool selection, spindle speed, and step-over need to be chosen carefully to balance accuracy against machine time. For moulds, 3-axis machining is used because it provides high accuracy and good surface finish, which is important for producing high-quality plastic parts. <Appendix E>



Figure 5.2 Illustration of X-, Y-, and Z-axis movements used for subtractive machining of mould components.

Conclusion

This report presented the full engineering design, analysis, and machining workflow for producing a moulded hair dryer housing using PTC Creo. The hair dryer was modelled with attention to ease of manufacture, including attention to wall thickness, draft angles, and internal webs. Once the part model was completed, a mould was created by assembling the reference geometry, adding a suitable parting surface, and adding runners and gates. Mould flow analysis showed issues with mouldability, air traps, and unbalanced filling, which were improved significantly by adding a third gate and adjusting the melt temperature.

The machining stage involved preparing the mould half for CNC manufacture by defining the stock, setting the coordinate system, and refining the roughing and finishing toolpaths. Several versions were required to achieve an accurate result, but the final toolpath produced a clean and viable mould cavity. Overall, the project demonstrated a complete and realistic manufacturing workflow from concept to machining.

Word Count (excluding title, contents, appendix, tables, captions, etc.): approx. 3900 words.

Appendix

Appendix A – Injection Moulding Best Practices (Source)

Crescent Industries (2023). *How to Ensure High-Quality Injection Molding: 6 Best Practices*. Available at: <https://www.crescentind.com/blog/how-to-ensure-high-quality-injection-molding-6-best-practices>

Appendix B – CNC G-Code (First 40 Lines)

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N5 G71
N10 ( / MANUFACTURING_HAIR_DRYER)
N15 G0 G17 G99
N20 G90 G94
N25 G0 G49
N30 T1 M06
N35 S1200 M03
N40 G0 G43 Z15. H1
N45 X152. Y-302.002
N50 Z10.4
N55 G1 Z4.6 F300.
N60 G3 X150. Y-300.002 I-2. J0.
N65 G1 X-.006
N70 X0. Y-299.844
N75 Y-.156
N80 X-.006 Y.002
N85 X300.006
N90 X300. Y-.156
N95 Y-299.844
N100 X300.006 Y-300.002
N105 X150.
N110 G2 X148.51 Y-299.142 I0. J1.72
N115 G1 X148.349 Y-298.862
N120 G3 X146.859 Y-298.002 I-1.49 J-.86
N125 G1 X2.
N130 Y-1.998
N135 X298.
N140 Y-298.002
N145 X150.
N150 G2 X148.51 Y-297.142 I0. J1.72
N155 G1 X148.349 Y-296.862
N160 G3 X146.859 Y-296.002 I-1.49 J-.86
N165 G1 X4.
N170 Y-3.998
N175 X296.
N180 Y-296.002
N185 X150.
N190 G2 X148.51 Y-295.142 I0. J1.72
N195 G1 X148.349 Y-294.862
N190 X135.902 Y-293.667 Z-10.933
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Appendix C – Injection Moulding Video Reference

YouTube (2023). *How a Hair Dryer Works – Behind the Scenes Manufacturing.*

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Appendix D – Injection Moulding Machine Anatomy

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Appendix E – CNC Machine Information

Mantool (2022). *3-Axis CNC Machining Centres – Overview and Applications.*

Available at: <https://mantool.com/3-axis-cnc-machining-centers/>

Appendix F – Plastic Shrinkage Rate Data

Upmold (2023). *Plastic Shrinkage Rate Chart.*

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